Mohan Krishna

Software & AI Engineer

BITS Pilani | B.E. Mathematics & Computing | Expected May 2028
Open to software and AI engineering roles and internships.

f20240732@pilani.bits-pilani.ac.in | +91 6302539619 | github.com/kirmada1509
linkedin.com/in/mohan-krishna-karthik | kirmada1509.github.io

Education

Birla Institute of Technology and Science (BITS), Pilani

B.E. Mathematics & Computing | Expected May 2028
Relevant coursework: Data Structures & Algorithms, Database Systems, Linear Algebra, Probability & Statistics, Optimization.

Experience

Engineering Lead | May 2026 - Present

Students' Union Technical Team, BITS Pilani

- Lead 20 student engineers across frontend, backend, mobile, and UI/UX to deliver campus products.
- Introduced sprint planning, code reviews, and engineering standards; helped increase daily active users by approximately 45%.

Software Engineering Intern | May - July 2026

OBS Experiences, Dehradun

- Architected a modular NestJS backend spanning 12+ business domains, including authentication, bookings, and payments.
- Shipped two production Next.js applications with shared components, reducing duplicate frontend code by 40%+.
- Automated deployment across four repositories with GitLab CI/CD, Docker, and Nginx, reducing deployment time by 90%.

Selected projects

IntervieHire - AI Interviewer | Sep 2026

LiveKit Agents, Fastify, WebSockets, PostgreSQL

- Migrated from Vapi to self-hosted LiveKit Agents with Deepgram STT, Silero VAD, and Cartesia TTS, reducing voice infrastructure cost by 58%+.
- Built bidirectional session orchestration with server-authoritative timing and idempotent interview turns.

ReEntry - Browser Task Agent | Oct 2026 | Technical preview

TypeScript, browser extension, Ollama / NVIDIA inference

- Built evidence-linked task recovery, checkpoint history, draft recovery, and conflict protection with provider-independent inference and user correction; documented evaluation and accessibility limits.

Precedent - Agent Action Checks | Sep 2026 | Prototype

Python, Moss, hybrid retrieval

- Built retrieval over approved and blocked tool calls using signatures, argument evidence, and provenance; added human review and live feedback write-back, with synthetic held-out evaluation.

OneTap Vendor | Dec 2025 - Mar 2026

Go, Fyne, SQLite, sqlc

- Built an offline-first campus credit billing system from scratch for 5,000+ students, supporting $1.2M+ in annual campus purchases.
- Vendors scan a QR in the SU app to charge the student's mess account, with billing at semester end; added retry/backoff synchronization and automatic backups.

BOSM - Sports Festival App | Aug - Sep 2025

React Native, Expo, TypeScript

- Developed schedules, leaderboards, verification, and food ordering for 6,000+ participants; owned iOS and Android production releases through Expo EAS, App Store, and Google Play.

Technical skills

Languages: Python, TypeScript/JavaScript, Go, C++, SQL
AI & real-time: LiveKit Agents, Deepgram STT, Cartesia TTS, Silero VAD, Ollama, NVIDIA inference, Moss, hybrid retrieval, evaluation
Applications: React, Next.js, React Native, NestJS, Fastify, FastAPI
Data & infrastructure: PostgreSQL, Redis, SQLite, Docker, GitLab CI/CD, Nginx

Achievements

- Top 10 campus finalist, Expedia Group Hackathon 2026: AI-powered multi-flight journey planner.
- LeetCode contest rating: 1700+.